

Measure Theory – 5

Product σ -Algebras and Hypographs, Product Measures and the
Fubini–Tonelli Theorems, Transition Kernels, Distribution Function
Identities, and Independence and Exchangeability

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Introduction

These notes collect the core definitions, results, and worked problems on product σ -algebras and measurability of hypographs, product measures and the Fubini–Tonelli theorems, measures built from transition kernels, the identity $\int F d\mu_F = \frac{1}{2}$ for continuous distribution functions, and independence and exchangeability of i.i.d. collections. The product-measure machinery is developed before the applications because the hypograph, kernel, and distribution-function problems all rest on Tonelli’s theorem.

Throughout, $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are measure spaces, λ denotes Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ (written μ_L in the assignment), and $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space.

1 Product σ -Algebras and Measurability

1.1 Definitions and Basic Properties

Definition 1.1 (Measurable rectangle; product σ -algebra). A **measurable rectangle** in $X \times Y$ is a set $A \times B$ with $A \in \mathcal{M}$, $B \in \mathcal{N}$. The **product σ -algebra** is

$$\mathcal{M} \otimes \mathcal{N} = \sigma(\{A \times B : A \in \mathcal{M}, B \in \mathcal{N}\}).$$

It is the smallest σ -algebra on $X \times Y$ making both coordinate projections π_X, π_Y measurable. (The assignment writes $\mathcal{M} \times \mathcal{N}$.)

The rectangles form a π -system, since $(A_1 \times B_1) \cap (A_2 \times B_2) = (A_1 \cap A_2) \times (B_1 \cap B_2)$. This is what makes the π - λ theorem applicable throughout the section.

Proposition 1.2 (Sections of measurable sets are measurable). *If $E \in \mathcal{M} \otimes \mathcal{N}$, then for every $x \in X$ the section $E_x = \{y \in Y : (x, y) \in E\}$ lies in $\mathcal{N}$. Consequently, if $f : X \times Y \rightarrow \mathbb{R}$ is $\mathcal{M} \otimes \mathcal{N}$ -measurable, then $y \mapsto f(x, y)$ is $\mathcal{N}$ -measurable for every x (and symmetrically in x).*

Proof. Let $\mathcal{G} = \{E \subseteq X \times Y : E_x \in \mathcal{N} \forall x\}$. It contains every rectangle, since $(A \times B)_x = B$ if $x \in A$ and $\emptyset$ otherwise. It is closed under complements, $(E^c)_x = (E_x)^c$, and countable unions, $(\bigcup_n E_n)_x = \bigcup_n (E_n)_x$. So $\mathcal{G}$ is a σ -algebra containing the rectangles, hence $\mathcal{M} \otimes \mathcal{N} \subseteq \mathcal{G}$. For the function statement, $\{y : f(x, y) \in C\} = (f^{-1}(C))_x$. $\square$

Proposition 1.3 (Maps into a product). *Let $h = (h_1, h_2) : (\Omega, \mathcal{F}) \rightarrow (X \times Y, \mathcal{M} \otimes \mathcal{N})$. Then h is measurable if and only if h_1 and h_2 are measurable. In particular, a map $h : Z \rightarrow \mathbb{R}^2$ is $(\mathcal{Z}, \mathcal{B}(\mathbb{R}^2))$ -measurable iff both coordinates are measurable, because $\mathcal{B}(\mathbb{R}^2) = \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$.*

Proof. $(\Rightarrow)$ $h_1 = \pi_X \circ h$ and $h_2 = \pi_Y \circ h$ are compositions of measurable maps. $(\Leftarrow)$ For a rectangle, $h^{-1}(A \times B) = h_1^{-1}(A) \cap h_2^{-1}(B) \in \mathcal{F}$. Since rectangles generate $\mathcal{M} \otimes \mathcal{N}$ and the collection of sets with measurable preimage is a σ -algebra, h is measurable. For $\mathbb{R}^2$, open rectangles $(a, b) \times (c, d)$ generate both $\mathcal{B}(\mathbb{R}^2)$ (every open set is a countable union of them) and $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$. $\square$

Common pitfalls.

- Sections of measurable sets are measurable, but the converse fails: a set all of whose sections are measurable need not lie in $\mathcal{M} \otimes \mathcal{N}$.
- $\mathcal{B}(\mathbb{R}^n) = \mathcal{B}(\mathbb{R})^{\otimes n}$ relies on $\mathbb{R}$ being second countable; for general spaces the Borel σ -algebra of the product can be strictly larger.

1.2 Problem: Measurability of the Under-Graph Region

Let $f : X \rightarrow [0, \infty)$ be $\mathcal{M}$ -measurable, and define the **hypograph**

$$A_f = \{(x, y) \in X \times \mathbb{R} : 0 \leq y \leq f(x)\}.$$

Claim: $A_f \in \mathcal{M} \otimes \mathcal{B}(\mathbb{R})$.

Proof. Define $h : X \times \mathbb{R} \rightarrow \mathbb{R}^2$ by $h(x, y) = (f(x) - y, y)$.

1. **h is measurable.** The map $(x, y) \mapsto f(x) = f(\pi_X(x, y))$ is $\mathcal{M} \otimes \mathcal{B}(\mathbb{R})$ -measurable as a composition of π_X and f , and $(x, y) \mapsto y = \pi_{\mathbb{R}}(x, y)$ is measurable. A difference of real measurable functions is measurable, so both coordinates of h are measurable, and by Proposition 1.3 h is $(\mathcal{M} \otimes \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}^2))$ -measurable.
2. **A_f is a preimage.** $h(x, y) \in [0, \infty) \times [0, \infty)$ iff $f(x) - y \geq 0$ and $y \geq 0$, i.e. iff $0 \leq y \leq f(x)$. Hence $A_f = h^{-1}([0, \infty) \times [0, \infty))$.
3. **Conclusion.** $[0, \infty)^2$ is closed, hence in $\mathcal{B}(\mathbb{R}^2)$, so $A_f \in \mathcal{M} \otimes \mathcal{B}(\mathbb{R})$. $\square$

Remark. The same argument shows that the strict hypograph $\{0 \leq y < f(x)\}$, the epigraph, and the graph $\{y = f(x)\} = h^{-1}(\{0\} \times \mathbb{R}) \cap \{y \geq 0\}$ are all in $\mathcal{M} \otimes \mathcal{B}(\mathbb{R})$; only the shape of the Borel set in $\mathbb{R}^2$ changes.

2 Product Measures and the Fubini–Tonelli Theorems

2.1 Construction of the Product Measure

Theorem 2.1 (Product measure). *If μ and ν are σ -finite, there is a unique measure $\mu \times \nu$ on $\mathcal{M} \otimes \mathcal{N}$ with*

$$(\mu \times \nu)(A \times B) = \mu(A) \nu(B) \quad (A \in \mathcal{M}, B \in \mathcal{N}).$$

Moreover, for every $E \in \mathcal{M} \otimes \mathcal{N}$ the functions $x \mapsto \nu(E_x)$ and $y \mapsto \mu(E^y)$ are measurable and

$$(\mu \times \nu)(E) = \int_X \nu(E_x) d\mu(x) = \int_Y \mu(E^y) d\nu(y),$$

where E_x and E^y denote the section at x and at y .

Proof (outline). First let μ, ν be finite. Let $\mathcal{D}$ be the class of $E \in \mathcal{M} \otimes \mathcal{N}$ for which $x \mapsto \nu(E_x)$ is measurable. It contains all rectangles $(\nu((A \times B)_x) = \mathbf{1}_A(x)\nu(B))$, is closed under proper differences $(\nu((E \setminus E')_x) = \nu(E_x) - \nu(E'_x))$ and increasing unions (continuity from below), so it is a λ -system; by the π - λ theorem $\mathcal{D} = \mathcal{M} \otimes \mathcal{N}$. Then $E \mapsto \int \nu(E_x) d\mu$ is a measure (countable additivity follows from MCT applied to $\sum_n \nu((E_n)_x)$) which equals $\mu(A)\nu(B)$ on rectangles. Uniqueness on the π -system of rectangles gives $\mu \times \nu$; the same argument in the other variable gives the second formula by uniqueness. For σ -finite measures, exhaust X and Y by increasing sequences of finite-measure rectangles, apply the finite case, and pass to the limit with continuity from below. $\square$

Theorem 2.2 (Tonelli). *Let μ, ν be σ -finite and $f : X \times Y \rightarrow [0, \infty]$ be $\mathcal{M} \otimes \mathcal{N}$ -measurable. Then $x \mapsto \int f(x, y) d\nu(y)$ and $y \mapsto \int f(x, y) d\mu(x)$ are measurable, and*

$$\int_{X \times Y} f d(\mu \times \nu) = \int_X \left(\int_Y f(x, y) d\nu(y) \right) d\mu(x) = \int_Y \left(\int_X f(x, y) d\mu(x) \right) d\nu(y),$$

with both sides possibly $+\infty$.

Theorem 2.3 (Fubini). *Let μ, ν be σ -finite and $f \in L^1(\mu \times \nu)$. Then $f(x, \cdot) \in L^1(\nu)$ for μ -a.e. x , $f(\cdot, y) \in L^1(\mu)$ for ν -a.e. y , the a.e.-defined functions $x \mapsto \int f(x, y) d\nu(y)$ and $y \mapsto \int f(x, y) d\mu(x)$ are integrable, and the same chain of equalities as in Tonelli holds.*

Proof (of Tonelli and Fubini from Theorem 2.1). For $f = \mathbf{1}_E$ the statement is Theorem 2.1. It extends to simple functions by linearity and to $f \geq 0$ by MCT along simple $s_k \uparrow f$ (Lemma on approximation), giving Tonelli. For Fubini, apply Tonelli to f^+ and f^- ; both have finite integral, so their inner integrals are finite a.e., and subtracting gives the result. $\square$

How to use the two theorems.

- **Tonelli** needs only $f \geq 0$ and σ -finiteness; use it to swap integrals freely and to *test integrability* by computing $\int |f| d(\mu \times \nu)$ as an iterated integral.
- **Fubini** needs $f \in L^1(\mu \times \nu)$; use it only after that has been checked via Tonelli on $|f|$.

2.2 Counterexamples: What Can Go Wrong

Example 2.4 (Iterated integrals that differ, integrability fails). On $(0, 1)^2$ with Lebesgue measure, let $f(x, y) = \frac{x^2 - y^2}{(x^2 + y^2)^2}$. Since $\frac{\partial}{\partial y} \frac{y}{x^2 + y^2} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$,

$$\int_0^1 \left(\int_0^1 f dy \right) dx = \int_0^1 \frac{dx}{1 + x^2} = \frac{\pi}{4}, \quad \int_0^1 \left(\int_0^1 f dx \right) dy = -\frac{\pi}{4}.$$

Here $\int |f| = \infty$, so Fubini does not apply.

Example 2.5 (σ -finiteness is needed). Let λ be Lebesgue measure and τ counting measure on $[0, 1]$ (τ is not σ -finite), and $D = \{(x, x)\}$ the diagonal, which is in $\mathcal{B} \otimes \mathcal{B}$. Then

$$\int \left(\int \mathbf{1}_D d\tau(y) \right) d\lambda(x) = \int 1 d\lambda = 1, \quad \int \left(\int \mathbf{1}_D d\lambda(x) \right) d\tau(y) = \int 0 d\tau = 0.$$

2.3 Problem: Area Under a Function Equals Its Integral

Let μ be σ -finite and $f : X \rightarrow [0, \infty)$ measurable, so $A_f \in \mathcal{M} \otimes \mathcal{B}(\mathbb{R})$ by Section 1. **Claim:**

$$(\mu \times \lambda)(A_f) = \int_X f d\mu.$$

Proof. By Tonelli,

$$(\mu \times \lambda)(A_f) = \int_X \left(\int_{\mathbb{R}} \mathbf{1}_{A_f}(x, y) d\lambda(y) \right) d\mu(x).$$

For fixed x the section is $(A_f)_x = [0, f(x)]$, so the inner integral is $\lambda([0, f(x)]) = f(x)$. Therefore $(\mu \times \lambda)(A_f) = \int_X f d\mu$. $\square$

Corollary 2.6 (Layer-cake formula). *Integrating in the other order, the section of A_f at height $y \geq 0$ is $\{x : f(x) \geq y\}$, so*

$$\int_X f d\mu = \int_0^\infty \mu(\{f \geq y\}) dy.$$

For a non-negative random variable, $\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X \geq y) dy = \int_0^\infty \mathbb{P}(X > y) dy$ (the two integrands differ for at most countably many y).

Example 2.7. • $\mu = \lambda$ on $[0, 1]$, $f(x) = x^2$: $\lambda(A_f) = \int_0^1 x^2 dx = \frac{1}{3}$.

- μ counting measure on $\{1, 2, 3\}$, $f = (1, 2, 3)$: A_f is a union of three vertical segments of lengths 1, 2, 3, and $(\mu \times \lambda)(A_f) = 6 = \sum f$.
- X a random variable with $\mathbb{P}(X \geq y) = e^{-y}$: $\mathbb{E}[X] = \int_0^\infty e^{-y} dy = 1$.

Remark. The graph $\{(x, y) : y = f(x)\}$ has $(\mu \times \lambda)$ -measure 0, since each section is a single point: $\int_X \lambda(\{f(x)\}) d\mu(x) = 0$. Hence open and closed versions of the region have the same area.

Common pitfalls.

- Measurability of A_f in the *product* σ -algebra must be established before Tonelli can be applied; that is the content of the hypograph problem.
- σ -finiteness of μ is a hypothesis of Tonelli; it is automatic for probability measures and Lebesgue measure.

3 Transition Kernels and Dissected Product Measures

3.1 Measures Built from a Marginal and a Kernel

Definition 3.1 (Transition kernel). A (**probability**) **transition kernel** from $(X_1, \mathcal{M}_1)$ to $(X_2, \mathcal{M}_2)$ is a map $\kappa : X_1 \times \mathcal{M}_2 \rightarrow [0, 1]$ such that

- for each $x_1 \in X_1$, $A_2 \mapsto \kappa(x_1, A_2) =: \mu_{x_1}(A_2)$ is a probability measure on $(X_2, \mathcal{M}_2)$;
- for each $A_2 \in \mathcal{M}_2$, $x_1 \mapsto \kappa(x_1, A_2)$ is $\mathcal{M}_1$ -measurable.

Theorem 3.2 (Dissection of a joint measure). *Let μ_1 be a probability measure on $(X_1, \mathcal{M}_1)$ and κ a kernel as above. Then*

$$\mu(A) = \int_{X_1} \mu_{x_1}(A_{x_1}) d\mu_1(x_1), \quad A \in \mathcal{M}_1 \otimes \mathcal{M}_2,$$

where $A_{x_1} = \{x_2 \in X_2 : (x_1, x_2) \in A\}$, defines a probability measure on $(X_1 \times X_2, \mathcal{M}_1 \otimes \mathcal{M}_2)$ with

$$\mu(A_1 \times A_2) = \int_{A_1} \mu_{x_1}(A_2) d\mu_1(x_1), \quad \mu(A_1 \times X_2) = \mu_1(A_1).$$

Proof. Well defined. By Proposition 1.2, $A_{x_1} \in \mathcal{M}_2$. The class of A for which $x_1 \mapsto \mu_{x_1}(A_{x_1})$ is measurable contains rectangles $(= \mathbf{1}_{A_1}(x_1)\mu_{x_1}(A_2))$ and is a λ -system (proper differences and increasing unions, as in the proof of Theorem 2.1), hence equals $\mathcal{M}_1 \otimes \mathcal{M}_2$. *Measure.* $\mu(\emptyset) = 0$ and $\mu(X_1 \times X_2) = \int 1 d\mu_1 = 1$. If $A^{(n)}$ are disjoint, then so are their sections, so $\mu_{x_1}((\bigcup_n A^{(n)})_{x_1}) = \sum_n \mu_{x_1}(A_{x_1}^{(n)})$, and MCT lets the sum pass through $\int \cdot d\mu_1$. The displayed formulas follow by substituting rectangles. $\square$

Special cases and integration.

- **Product measure.** If $\mu_{x_1} = \mu_2$ for all x_1 , then $\mu(A) = \int \mu_2(A_{x_1}) d\mu_1 = (\mu_1 \times \mu_2)(A)$.
- **Integration against μ .** For $f \geq 0$ measurable (or $f \in L^1(\mu)$),

$$\int f d\mu = \int_{X_1} \left(\int_{X_2} f(x_1, x_2) d\mu_{x_1}(x_2) \right) d\mu_1(x_1),$$

proved by the same ladder from indicators to general f .

- **Marginals.** The first marginal is μ_1 ; the second is $A_2 \mapsto \int \mu_{x_1}(A_2) d\mu_1(x_1)$, a μ_1 -mixture of the kernel measures.

Example 3.3 (A two-level model). Let $X_1 = \{0, 1\}$, $\mu_1(\{1\}) = p$, and let $\mu_0 = N(0, 1)$, $\mu'_1 = N(m, 1)$ be the kernel measures at $x_1 = 0, 1$. Then the joint measure of (x_1, x_2) is $\mu(\{x_1\} \times B) = \mu_1(\{x_1\})\mu_{x_1}(B)$, and the marginal of x_2 is the mixture $(1 - p)N(0, 1) + pN(m, 1)$. The joint is a product measure iff $m = 0$ (kernel constant in x_1).

Common pitfalls.

- Measurability of $x_1 \mapsto \mu_{x_1}(A_2)$ is part of the definition of a kernel; without it the integral defining μ need not make sense.
- A joint measure with given marginals is not unique; the kernel encodes the dependence between the coordinates.

4 Distribution Function Identities

4.1 Problem: $\int F d\mu_F = \frac{1}{2}$ for Continuous F

Let F be a continuous distribution function with probability measure μ_F (Lebesgue–Stieltjes).

Claim:

$$\int_{\mathbb{R}} F(x) d\mu_F(x) = \frac{1}{2}.$$

Proof. Let $A = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \leq x_2\}$; it is closed, so $A \in \mathcal{B}(\mathbb{R}^2) = \mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$. Both measures are finite, so Tonelli applies on $(\mathbb{R}^2, \mu_F \times \mu_F)$. Integrating first over x_2 for fixed x_1 , the section is $[x_1, \infty)$ with $\mu_F([x_1, \infty)) = 1 - F(x_1^-) = 1 - F(x_1)$ by continuity, so

$$(\mu_F \times \mu_F)(A) = \int_{\mathbb{R}} \mu_F([x_1, \infty)) d\mu_F(x_1) = \int_{\mathbb{R}} (1 - F(x_1)) d\mu_F(x_1) = 1 - \int_{\mathbb{R}} F d\mu_F.$$

Now use symmetry. The swap $\sigma(x_1, x_2) = (x_2, x_1)$ preserves $\mu_F \times \mu_F$ (the push-forward agrees on rectangles, hence everywhere by uniqueness), and $\sigma^{-1}(A) = A' = \{x_1 \geq x_2\}$. So $(\mu_F \times \mu_F)(A) = (\mu_F \times \mu_F)(A')$. Also $A \cup A' = \mathbb{R}^2$ and $A \cap A' = D = \{x_1 = x_2\}$, and by Tonelli and continuity

$$(\mu_F \times \mu_F)(D) = \int_{\mathbb{R}} \mu_F(\{x_1\}) d\mu_F(x_1) = 0,$$

since F continuous means μ_F has no atoms. Hence $2(\mu_F \times \mu_F)(A) = (\mu_F \times \mu_F)(A \cup A') + (\mu_F \times \mu_F)(D) = 1$, so $(\mu_F \times \mu_F)(A) = \frac{1}{2}$ and therefore $1 - \int F d\mu_F = \frac{1}{2}$, i.e. $\int F d\mu_F = \frac{1}{2}$. $\square$

Remark. • **Probabilistic reading.** If X_1, X_2 are i.i.d. with law μ_F , then $\int F d\mu_F = \mathbb{P}(X_2 \leq X_1) = \frac{1}{2}$: for a continuous distribution each of two independent draws is equally likely to be the larger.

• **Second proof.** For continuous F , $F(X) \sim \text{Uniform}(0, 1)$, so $\int F d\mu_F = \mathbb{E}[F(X)] = \frac{1}{2}$ (change of variables).

4.2 What Happens Without Continuity

Suppose F is any distribution function. The same argument gives $(\mu_F \times \mu_F)(A') = \int F d\mu_F$ and, by symmetry, the two measures are equal, so

$$2 \int_{\mathbb{R}} F d\mu_F = (\mu_F \times \mu_F)(\mathbb{R}^2) + (\mu_F \times \mu_F)(D) = 1 + \sum_p \mu_F(\{p\})^2,$$

where D has measure $\int \mu_F(\{x\}) d\mu_F(x) = \sum_p \mu_F(\{p\})^2$ over the atoms p . Thus

$$\int_{\mathbb{R}} F d\mu_F = \frac{1}{2} \left(1 + \sum_p \mu_F(\{p\})^2 \right).$$

Example 4.1. Let $\mu_F = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$. Then $F(0) = \frac{1}{2}$, $F(1) = 1$, and

$$\int F d\mu_F = \frac{1}{2} \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{3}{4} = \frac{1}{2} \left(1 + \frac{1}{4} + \frac{1}{4} \right),$$

which is not $\frac{1}{2}$: continuity is needed, because ties $X_1 = X_2$ now have positive probability.

Common pitfalls.

- Continuity is used twice: to get $\mu_F([x_1, \infty)) = 1 - F(x_1)$ (rather than $1 - F(x_1^-)$) and to make the diagonal null.
- The diagonal must be shown to be null via Tonelli; it is a Lebesgue-null set in $\mathbb{R}^2$, but $\mu_F \times \mu_F$ need not be absolutely continuous with respect to Lebesgue measure.

5 Independence and Exchangeability

5.1 Independence via Product Measures

Definition 5.1 (Independence). Random variables $X_1, \dots, X_m$ on $(\Omega, \mathcal{F}, \mathbb{P})$ are **independent** if

$$\mathbb{P}(X_1 \in B_1, \dots, X_m \in B_m) = \prod_{i=1}^m \mathbb{P}(X_i \in B_i) \quad \text{for all Borel } B_i.$$

They are **i.i.d.** if in addition they share a common law.

Proposition 5.2 (Characterizations). *The following are equivalent:*

- (a) $X_1, \dots, X_m$ are independent;
- (b) the joint law of $(X_1, \dots, X_m)$ is the product measure $\mathbb{P}_{X_1} \times \dots \times \mathbb{P}_{X_m}$ on $\mathcal{B}(\mathbb{R}^m)$;
- (c) the joint distribution function factorizes: $F_{X_1, \dots, X_m}(x_1, \dots, x_m) = \prod_i F_{X_i}(x_i)$.

Proof. (a) $\iff$ (b): the two probability measures on $\mathbb{R}^m$ agree on all Borel rectangles by (a) (and by definition of the product measure), which form a π -system generating $\mathcal{B}(\mathbb{R}^m) = \mathcal{B}(\mathbb{R})^{\otimes m}$; conversely, evaluate the product measure on a rectangle. (b) $\implies$ (c): evaluate on $\prod_i (-\infty, x_i]$. (c) $\implies$ (b): the sets $\prod_i (-\infty, x_i]$ form a π -system generating $\mathcal{B}(\mathbb{R}^m)$ on which the two probability measures agree, so they agree everywhere by the uniqueness theorem. $\square$

Proposition 5.3 (Product of independent integrable variables). *If X, Y are independent and integrable, then XY is integrable and $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.*

Proof. By change of variables and Proposition 5.2, $\mathbb{E}|XY| = \int |xy| d(\mathbb{P}_X \times \mathbb{P}_Y)$, which by Tonelli equals $\mathbb{E}|X|\mathbb{E}|Y| < \infty$. Fubini then gives $\mathbb{E}[XY] = \int x \left(\int y d\mathbb{P}_Y \right) d\mathbb{P}_X = \mathbb{E}[X]\mathbb{E}[Y]$. $\square$

5.2 Problem: Exchangeability of i.i.d. Collections

Definition 5.4 (Exchangeability). $X_1, \dots, X_m$ are **exchangeable** if for every permutation $(i_1, \dots, i_m)$ of $(1, \dots, m)$, the vectors $(X_{i_1}, \dots, X_{i_m})$ and $(X_1, \dots, X_m)$ have the same joint law.

Claim: if $X_1, \dots, X_m$ are i.i.d., they are exchangeable.

Proof. Write F_X for the common distribution function. For any permutation $(i_1, \dots, i_m)$ and $x_1, \dots, x_m \in \mathbb{R}$,

$$F_{X_{i_1}, \dots, X_{i_m}}(x_1, \dots, x_m) = \prod_{j=1}^m \mathbb{P}(X_{i_j} \leq x_j) = \prod_{j=1}^m F_X(x_j) = F_{X_1, \dots, X_m}(x_1, \dots, x_m).$$

The first equality is independence of the (re-indexed) family, the second is identical distribution. Since a joint distribution function determines the joint law (Proposition 5.2, via the π -system of orthants), the two vectors have the same law. $\square$

Remark. • Exchangeability is strictly weaker than i.i.d. Take $X_1 = X_2 = Z$ with $Z \sim \text{Bernoulli}(\frac{1}{2})$: the pair is exchangeable, but not independent. Sampling without replacement from an urn is another example.

- Exchangeability is a statement about the *joint* law of the vector, not merely about identical marginals.

Corollary 5.5 (Ranks and ties for i.i.d. continuous samples). *Let $X_1, \dots, X_m$ be i.i.d. with continuous distribution function. Then ties have probability zero and every ordering is equally likely:*

$$\mathbb{P}(X_{\pi(1)} < X_{\pi(2)} < \dots < X_{\pi(m)}) = \frac{1}{m!} \quad \text{for each permutation } \pi.$$

Proof. For $i \neq j$, $\mathbb{P}(X_i = X_j) = (\mu_F \times \mu_F)(D) = 0$ as in Section 4, so a.s. all values are distinct. The $m!$ events $\{X_{\pi(1)} < \dots < X_{\pi(m)}\}$ are disjoint, cover a set of probability 1, and have equal probability by exchangeability. $\square$

Remark. For $m = 2$ this recovers the identity of Section 4: $\mathbb{P}(X_2 \leq X_1) = \int F d\mu_F = \frac{1}{2}$.

Common pitfalls.

- The factorization of the joint distribution function is equivalent to independence, but the same product formula for marginals alone (without independence) proves nothing about the joint law.
- Continuity of the common law is essential for the $1/m!$ conclusion; with atoms, ties occur with positive probability.